WORKSHEET 8.5

Area Between Curves, Slicing Horizontally

Unit 8: Applications of Integration

Key ideas

- When the boundaries are given as $x = \ldots$ in terms of y, slice horizontally: area $= \displaystyle\int_c^d (\text{right} - \text{left})\, dy$.
- The limits are y-values.
- A curve $y = f(x)$ can be rewritten as $x = \ldots$ when that is easier.
- Choose the direction of slicing that needs only one integral.

Worked examples

Example 1 · A parabola and a line

Find the area enclosed by $x = y^2$ and $x = 2y + 3$.

Solution. $y^2 - 2y - 3 = 0$ gives $y = -1, 3$. The line is on the right: $\displaystyle\int_{-1}^{3} (2y + 3 - y^2)\, dy$.

$\dfrac{32}{3}$.

Example 2 · A curve and the y-axis

Find the area enclosed by $x = 9 - y^2$ and the y-axis.

Solution. $\displaystyle\int_{-3}^{3} (9 - y^2)\, dy = 54 - 18$.

36.

Example 3 · Rewriting a curve

Find the area of the region bounded by $y = \ln x$, the y-axis, and the lines $y = 0$ and $y = 1$.

Solution. $x = e^y$ is the right boundary: $\displaystyle\int_0^1 e^y\, dy$.

$e - 1$.

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Practice

CORE

1. Rewrite $y = x^3$ in the form $x = \ldots$

2. Write, but do not evaluate, an integral in y for the area enclosed by $x = y^2$ and $x = 4$.

3. Evaluate the integral in Question 2.

4. Find the area enclosed by $x = y^2$ and $x = y$.

EXAM LEVEL

5. Find the area enclosed by $x = y^2 - 5$ and $x = 3 - y^2$.

6. Find the area of the region bounded by $y = \sqrt{x}$, the y-axis and the line $y = 3$.

7. Find the area of the region in the first quadrant enclosed by $x = y^3$ and $x = 4y$.

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8. Find the area enclosed by $x = y^2 - 2y$ and the y-axis.

Question 9 · Free-response practice

Let R be the region enclosed by the parabola $x = y^2$ and the line $x = 8 - 2y$.

(a) Find the coordinates of the two points of intersection.

(b) Find the area of R.

(c) Write, but do not evaluate, an expression involving integrals with respect to x that gives the area of R.

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Worked solutions

1. $x = \sqrt[3]{y}$.

2. $\displaystyle\int_{-2}^{2} (4 - y^2)\,dy$.

3. $\dfrac{32}{3}$.

4. $\displaystyle\int_{0}^{1} (y - y^2)\,dy = \frac{1}{6}$.

5. They meet at $y = \pm 2$: $\displaystyle\int_{-2}^{2} (8 - 2y^2)\,dy = \frac{64}{3}$.

6. $x = y^2$: $\displaystyle\int_{0}^{3} y^2\,dy = 9$.

7. They meet at $y = 0$ and $y = 2$: $\displaystyle\int_{0}^{2} (4y - y^3)\,dy = 4$.

8. The curve lies to the left of the axis for $0 < y < 2$: $\displaystyle\int_{0}^{2} (2y - y^2)\,dy = \frac{4}{3}$.

9. (a) $y^2 + 2y - 8 = 0$ gives $y = -4$ and $y = 2$: the points $(16, -4)$ and $(4, 2)$.

   (b) $\displaystyle\int_{-4}^{2} (8 - 2y - y^2)\,dy = 36$.

   (c) $\displaystyle\int_{0}^{4} 2\sqrt{x}\,dx + \int_{4}^{16} \left(\frac{8 - x}{2} + \sqrt{x}\right) dx$.

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